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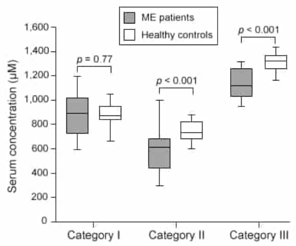

Figure 1 Serum concentrations of amino acids in ME patients and healthy controls

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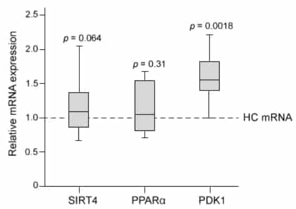


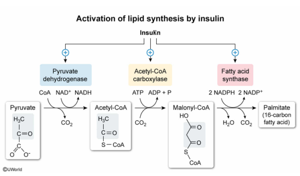
Figure 2 Gene expression of SIRT4, PPARα, and PDK1 in PBMCs from ME patients. The dashed line shows the relative expression levels of each gene in healthy control (HC) cells. The given *p*-values are relative to HC cells.

Lipids serve as energy stores for healthy patients and patients with ME. If both groups were fed the same diet, would the rate of lipid synthesis after a meal be faster in healthy individuals or in patients with ME?

- A. Patients with ME, because the release of glucagon activates acetyl-CoA carboxylase in the liver [11%]
- B. Patients with ME, because more chylomicrons are broken down in the bloodstream [13%]
- ✔ C. Healthy individuals, because insulin can stimulate the transport of more acetyl-CoA to the cytosol [55%]
- D. Healthy individuals, because fewer fatty acids are being generated by hormone-sensitive lipase [19%]

Correct
55% answered correctly

Explanation:



The reactions involved in lipid synthesis are sensitive to hormones such as insulin and glucagon that control blood sugar (glucose) levels. Insulin reduces blood sugar levels by increasing glucose uptake and storage, whereas glucagon increases the production of glucose and its release into the bloodstream. Ongoing digestion after a meal leads to high blood sugar levels that induce the release of insulin and the suppression of glucagon production.

Liver cells respond to **insulin** by **increasing** their rates of lipid synthesis. In the mitochondria, insulin activates **pyruvate dehydrogenase (PDH)**, which converts pyruvate to acetyl-CoA. The accumulated acetyl-CoA is transported to the cytoplasm through the citrate shuttle, in which acetyl-CoA combines with oxaloacetate to form citrate. Citrate can then exit the mitochondria. In the cytoplasm, citrate is converted back to acetyl-CoA and oxaloacetate. Multiple acetyl-CoA units can then be combined to form fatty acyl groups.

According to the passage, PDH has a lower activity in patients with ME than in healthy controls. Consequently, even if insulin binds to liver cells after a meal, the pyruvate produced from glucose is not efficiently converted to acetyl-CoA and does not contribute efficiently to fatty acid synthesis. In contrast, healthy individuals would have a greater response to insulin because they have functional PDH that produces **higher quantities** of acetyl-CoA. Therefore, the transport of acetyl-CoA to the cytoplasm is stimulated in **healthy individuals**, who have a **faster rate of lipid synthesis**.

(Choice A) Glucagon *reduces* the rate of lipid synthesis by inhibiting acetyl-CoA carboxylase.

(Choice B) The digestion of chylomicrons (globules of protein and lipids) depends on the activity of lipoprotein lipase, which is not tested in the experiment.

(Choice D) Hormone-sensitive lipase converts triacylglycerides to fatty acids and glycerol. After a meal, insulin *inhibits* this enzyme in adipocytes (fat cells). Decreased hormone-sensitive lipase activity will *not* contribute to an increased rate of lipid synthesis.

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Educational objective:
Insulin stimulates lipid synthesis by activating the major enzymes involved in fatty acid production: pyruvate dehydrogenase complex, acetyl-CoA carboxylase, and fatty acid synthase. PDH converts pyruvate produced from glucose to acetyl-CoA, which is transported to the cytoplasm for fatty acid synthesis.

Time spent:0 sec
Subject:
Biochemistry

QID:401206
System:
Metabolic Reactions

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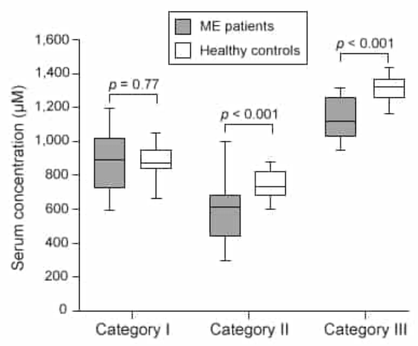

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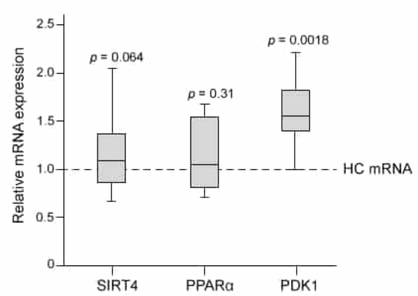

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If lower phospholipid levels in ME patients are due to inhibition of fatty acid synthesis, researchers will most likely observe which of the following?

✔ A. Increased cytosolic NADPH [49%]

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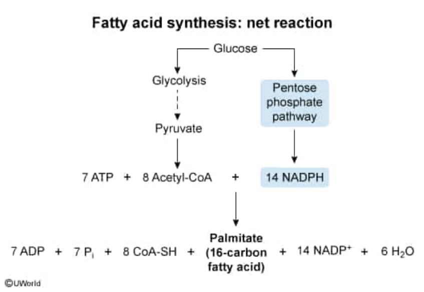
◯ B. Decreased mitochondrial acetyl-CoA [29%]
◯ C. Decreased mitochondrial fumarate [7%]
◯ D. Increased cytosolic lysine [12%]

Correct
49% answered correctly

Explanation:



Fatty acid synthesis is an anabolic process that **builds lipids in the cytosol**. Anabolic processes generally require energy, reducing power, and sufficient precursor molecules. In fatty acid synthesis, these requirements are satisfied by **ATP, NADPH, and acetyl-CoA**, respectively.

During fatty acid synthesis, NADPH is oxidized to $NADP^+$ to reduce the carbonyl groups and subsequent carbon-carbon double bonds on each acetyl-CoA molecule added to the fatty acid chain. If fatty acid synthesis is inhibited in ME patients, the conversion of NADPH to $NADP^+$ will occur less frequently, and cytosolic **NADPH will build up**.

(Choice B) Fatty acid synthesis takes place in the cytosol, not the mitochondria. In addition, cytosolic acetyl-CoA levels would most likely increase when fatty acid synthesis is inhibited because less acetyl-CoA is consumed.

(Choice C) Fumarate concentration is not directly related to fatty acid synthesis and would most likely be unaffected.

(Choice D) Lysine is not directly involved in fatty acid synthesis. In addition, Figure 1 shows that Category II amino acids, including lysine (K), had decreased levels in the plasma of ME patients and would likely be decreased in the cytosol as well.

Educational objective:
Acetyl-CoA, NADPH, and ATP are the reactants needed to generate fatty acid chains during lipid synthesis. The pentose phosphate pathway generates the NADPH needed to reduce the carbonyl groups from each molecule of acetyl-CoA that is added to a fatty acid chain.

Time spent:0 sec
Subject:
Biochemistry

QID:401207
System:
Metabolic Reactions

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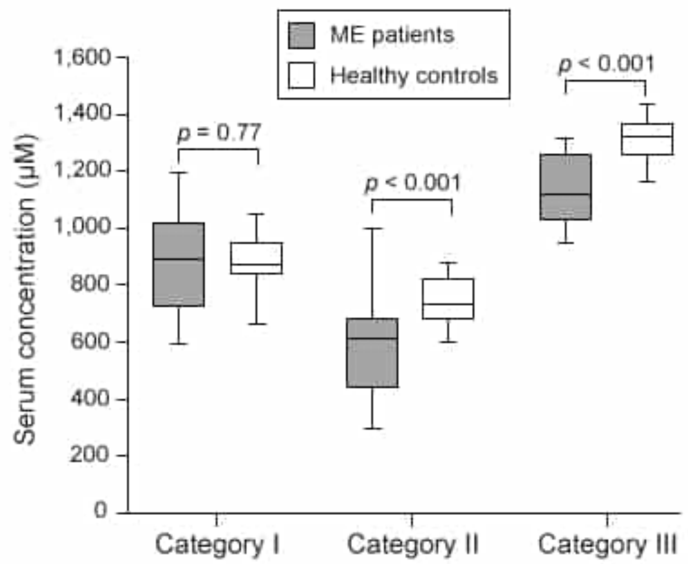

Figure 1 Serum concentrations of amino acids in ME patients and healthy controls

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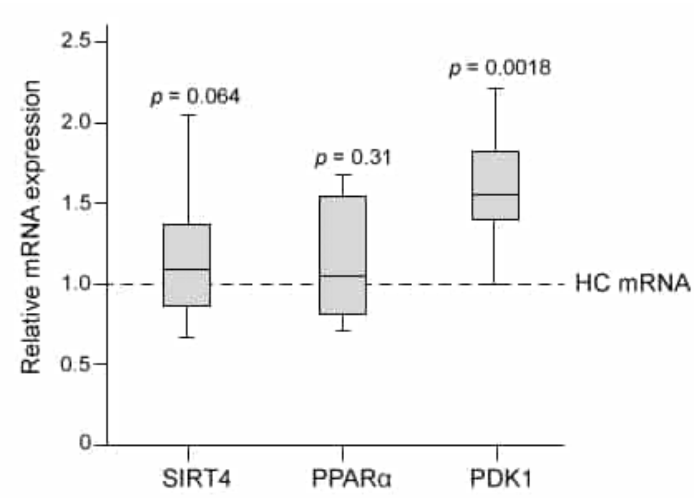

Figure 2 Gene expression of SIRT4, PPARα, and PDK1 in PBMCs from ME patients. The dashed line shows the relative expression levels of each gene in healthy control (HC) cells. The given *p*-values are relative to HC cells.

To confirm that β-oxidation was NOT impaired in healthy controls, the researchers most likely measured which metabolic step in PBMCs?

- A. The export of citrate to the cytoplasm to produce acetyl-CoA [11%]
- B. The release of CO_2 from malonyl-CoA [18%]
- C. The translocation of acylcarnitine to the mitochondrial matrix

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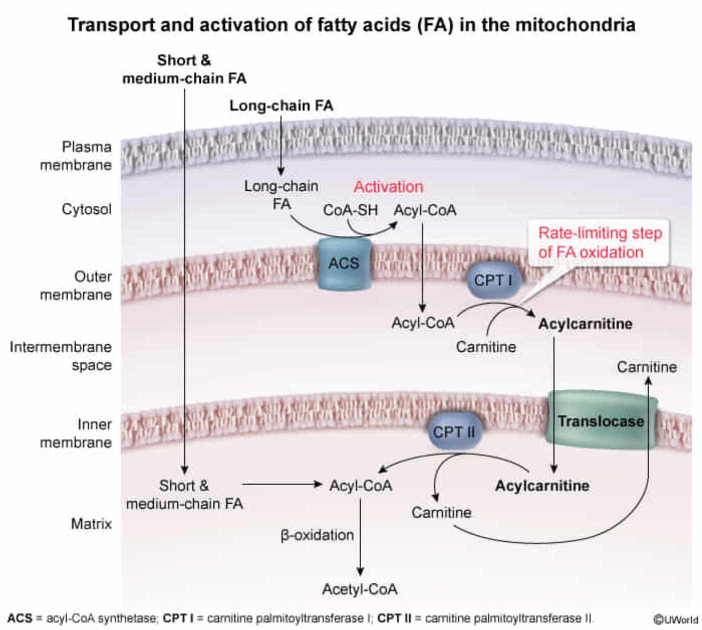
[61%]

D. The depletion of reduced electron carriers [8%]

Correct
62% answered correctly

Explanation:



β-oxidation is the breakdown of fatty acid chains into acetyl-CoA through a series of oxidation reactions in the mitochondrial matrix that produce NADH and $FADH_2$, which can be used to synthesize ATP. Short- and medium-chain fatty acids can diffuse through the inner and outer membranes of the mitochondria, but long-chain fatty acids obtained from lipid stores must undergo a set of reactions to enter the mitochondrial matrix.

First, fatty acids in the cytoplasm are **activated** when the enzyme acyl-

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CoA synthetase attaches them to the carrier molecule CoA using ATP. Subsequently, in the rate-limiting step of the reaction, **carnitine palmitoyltransferase I** (CPTI) converts **fatty acyl-CoA molecules to fatty acylcarnitine**, which enters the intermembrane space. Fatty acylcarnitine is then moved by **acylcarnitine translocase** across the inner membrane and into the matrix. Lastly, to begin the oxidation reactions, carnitine palmitoyltransferase II (CPTII) on the inner membrane reconverts fatty acylcarnitine to fatty acyl-CoA, which can be broken down into acetyl-CoA. Therefore, if the researchers wanted to confirm that fatty acid oxidation was not impaired, they would need to ensure that translocation of acylcarnitine into the mitochondrial matrix was functional.

(Choice A) During fatty acid synthesis, acetyl-CoA in the mitochondria is converted to citrate and then transported into the cytoplasm, where it is reconverted to acetyl-CoA and oxaloacetate. This does not occur during β-oxidation.

(Choice B) CO_2 is released from malonyl-CoA during fatty acid synthesis but not during fatty acid oxidation.

(Choice D) β-oxidation reactions transfer electrons from fatty acids to the electron carriers NAD^+ and FAD to produce their reduced forms. Therefore, reduced electron carriers are created, rather than depleted, by β-oxidation.

Educational objective:

During β-oxidation, long-chain fatty acids are activated with coenzyme A and shuttled by enzymes from the cytoplasm to the mitochondrial matrix. The rate-limiting step of fatty acid oxidation is the conversion of fatty acyl-CoA molecules to fatty acylcarnitine by carnitine palmitoyltransferase I.

Time spent:0 sec	**QID**:401208
Subject:	**System:**
Biochemistry	Metabolic Reactions

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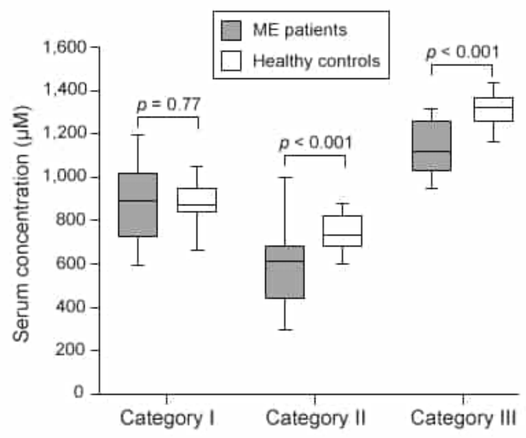

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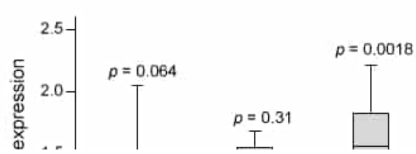

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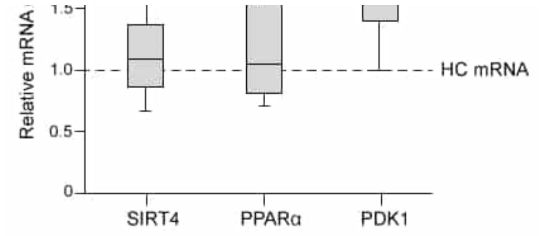

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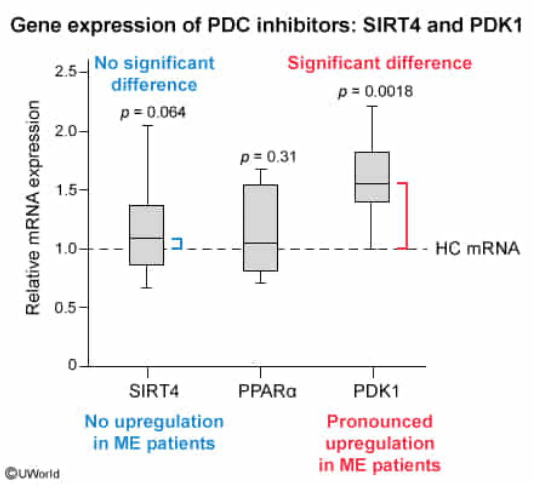
Based on Figure 2, the availability of ATP in PBMCs is most likely affected by which process?

- A. Overexpression of PDH mRNA [6%]
- B. Removal of lipoic acid cofactors by SIRT4 [5%]
- C. Upregulation of PPARα transcription factors [6%]
- D. Phosphorylation of PDH by PDK1 [81%]

Correct
81% answered correctly

Explanation:



The **availability of ATP** within a cell depends on the enzymes that control the production of intermediates used in **glycolysis**, the **citric acid cycle**, and **fatty acid oxidation**. The pyruvate dehydrogenase complex (PDC) links glycolysis to the citric acid cycle by catalyzing the conversion of **pyruvate to acetyl-CoA** in a decarboxylation reaction. This decarboxylation requires CoA, thiamine pyrophosphate, lipoic acid, and NAD^+ to remove a molecule of CO_2 from pyruvate and produce acetyl-CoA and NADH. The PDC is composed of pyruvate dehydrogenase (PDH), a transacetylase, and dihydrolipoyl dehydrogenase. It is normally regulated by two enzymes: pyruvate dehydrogenase kinase (PDK), and pyruvate dehydrogenase phosphatase (PDP).

Scientists predicted that ME patients might have increased expression of regulatory proteins such as SIRT4 and PDK1, which could decrease PDH activity. Of the proteins tested in the box-and-whisker plot shown in Figure 2, only PDK1 mRNA levels were significantly higher in patients with ME than in healthy controls (ie, only PDK1 has a ***p*-value that is less than 0.05** relative to

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healthy cells). Regulatory kinases such as PDK1 can alter the activity of their target proteins by adding phosphate groups (phosphorylation). Therefore, the decrease in ATP levels in patients with ME is most likely due to the phosphorylation of PDH by PDK1.

(Choice A) Expression levels of PDH mRNA were not tested.

(Choice B) SIRT4 can inhibit PDH activity by removing lipoic acid and therefore could affect ATP levels. However, Figure 2 shows that SIRT4 mRNA levels in patients with ME were not significantly different than those in healthy controls.

(Choice C) PPARα transcription factors are involved in fatty acid oxidation, which affects ATP levels. However, Figure 2 shows that PPARα mRNA levels were unaffected in patients with ME.

Educational objective:
The pyruvate dehydrogenase complex catalyzes the decarboxylation reaction that converts pyruvate to acetyl-CoA. This reaction requires CoA, thiamine pyrophosphate, lipoic acid, and NAD^+ and links glycolysis to the citric acid cycle. These pathways are essential for maximal ATP synthesis.

Time spent:0 sec
Subject:
Biochemistry

QID:401209
System:
Metabolic Reactions

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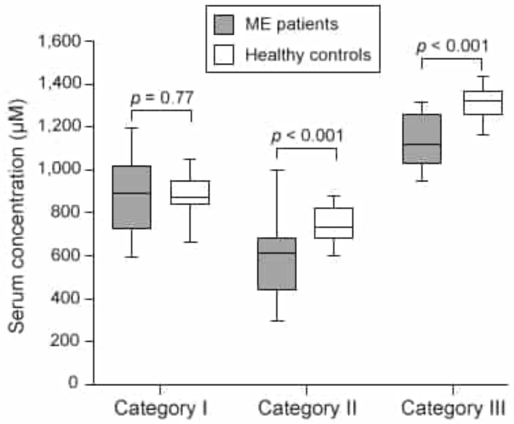

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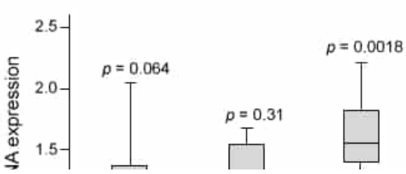

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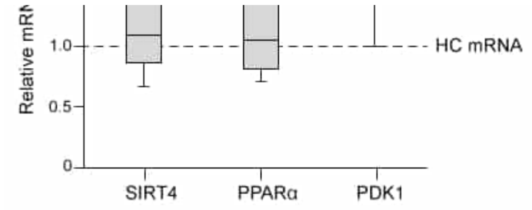

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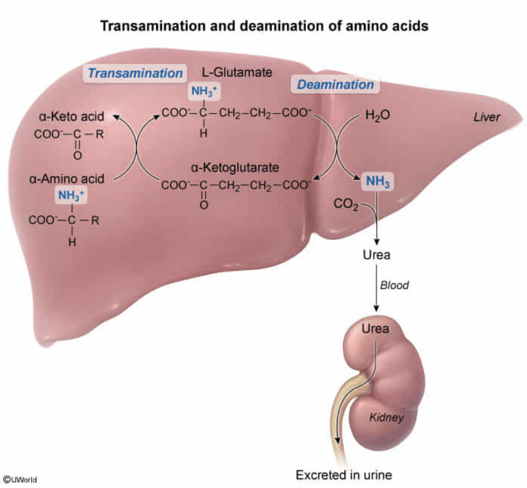
Assuming that 3-methylhistidine is the only indicator of protein catabolism, which observation would best support the difference in the metabolism of amino acids in male and female patients with ME?

- A. Male ME patients undergo fewer transamination reactions than female patients. [14%]
- B. Female ME patients excrete more ammonia in their urine than male patients. [14%]
- C. Male ME patients produce more glutamate in their liver cells than female patients. [56%]
- D. Female ME patients produce more α-keto acids in their muscles than male patients. [13%]

Correct
56% answered correctly

Explanation:



Catabolism is the breakdown of large molecules into smaller units and leads to production of energy in the form of ATP. **Protein catabolism** in the liver refers to the breakdown of polypeptide chains and proteins into individual amino acids, which can be further metabolized to enter the citric acid cycle and produce ATP. This requires a **transamination** reaction that removes the amino group ($-NH_3^+$) from the carbon backbone of the amino acid. The transamination reaction produces an α-keto acid from the amino acid by transferring the $-NH_3^+$ group to **α-ketoglutarate**, forming **glutamate.** The glutamate produced in this process is then deaminated, releasing ammonia (NH_3), which enters the urea cycle. The urea formed in this cycle is released in urine.

In the passage, 3-methylhistidine is described as an indicator of protein catabolism that is higher in male than in female ME patients. This indicates that proteins in male patients are catabolized

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at a faster rate than in female patients. As a result, male patients would be expected to produce a **higher concentration of glutamate** in their liver cells compared with female patients.

(Choice A) Male patients with ME undergo *more* transamination reactions than female patients because protein catabolism is increased.

(Choice B) The higher rate of protein catabolism in male patients indicates that their urine contains higher concentrations of ammonia than the urine of female patients.

(Choice D) α-Keto acids are more concentrated in male patients with ME than in female patients with ME because more transamination reactions occur in male patients.

Educational objective:
Protein catabolism refers to the breakdown of polypeptide chains and proteins into individual amino acids to produce ATP, glucose, or new proteins. Transamination reactions generate α-keto acids from amino acids by transferring the $-NH_3^+$ group to α-ketoglutarate, which is converted to glutamate.

Time spent:0 sec	**QID**:401210
Subject:	**System:**
Biochemistry	Metabolic Reactions

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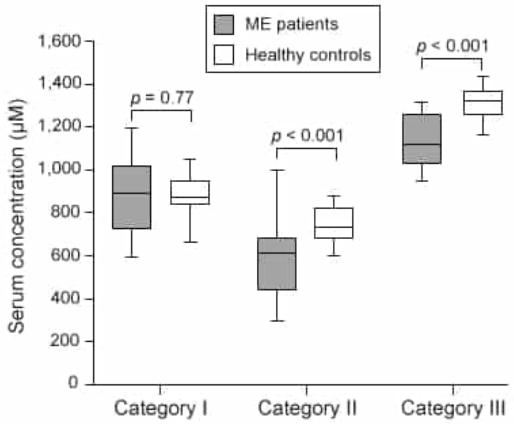

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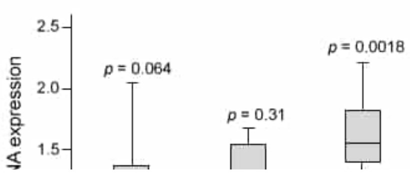

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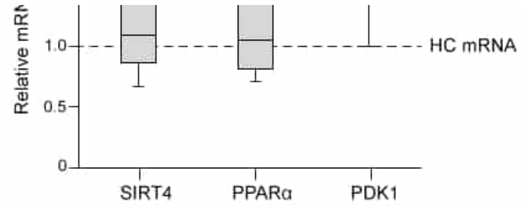


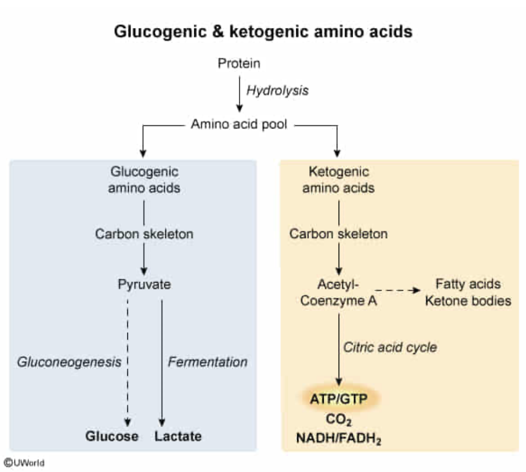
Figure 2 Gene expression of SIRT4, PPARα, and PDK1 in PBMCs from ME patients. The dashed line shows the relative expression levels of each gene in healthy control (HC) cells. The given *p*-values are relative to HC cells.

Extended and intense periods of physical exertion in patients with ME result in the conversion of glycine and cysteine to which of the following compounds in muscle cells?

- A. Fumarate [5%]
- B. Lactate [76%]
- C. Glucose [11%]
- D. Isocitrate [6%]

Correct
77% answered correctly

Explanation:



Amino acids are classified as glucogenic or ketogenic depending on the metabolic intermediates to which they are converted. **Glucogenic** amino acids are converted to pyruvate or citric acid cycle intermediates, which can then be converted to glucose. **Ketogenic** amino acids are converted directly to acetyl-CoA, which can enter the citric acid cycle or be used to form ketone bodies. The passage uses this classification to place amino acids into the three categories in Table 1. Amino acids in Category I are converted exclusively to pyruvate, and those in Category II become acetyl-CoA. Category III consists of amino acids used to make various citric acid cycle intermediates.

Glycine and cysteine are Category I amino acids, which are converted directly to pyruvate. Patients with ME at rest have sufficient ATP to convert pyruvate to citric acid cycle intermediates (via oxaloacetate) during aerobic respiration. However, the shortage of oxygen caused by prolonged physical exertion inhibits aerobic respiration by preventing NAD^+ and FAD, which are used by the citric acid cycle, from being regenerated.

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To maintain ATP production during extended and intense exercise, muscle cells switch from aerobic to anaerobic respiration, which does not require oxygen and consists of glycolysis and fermentation. Glycolysis contributes two molecules of ATP for each glucose molecule converted to pyruvate. During fermentation, the NAD^+ required for glycolysis to continue is generated by reducing pyruvate to lactate using the enzyme lactate dehydrogenase. Therefore, glycine and cysteine are ultimately converted to lactate during extended, intense periods of physical exertion.

(Choices A and D) Patients with ME cannot convert pyruvate to acetyl-CoA, which is a precursor molecule for citric acid cycle intermediates such as fumarate and isocitrate.

(Choice C) Conversion of pyruvate to glucose occurs in gluconeogenesis, which requires ATP. Aerobic respiration is required to produce sufficient ATP for gluconeogenesis.

Educational objective:
Glucogenic amino acids are amino acids that can enter gluconeogenesis. Ketogenic amino acids are converted directly to acetyl-CoA, which can enter the citric acid cycle or be used to form ketone bodies. In the absence of oxygen, pyruvate is reduced to lactate to regenerate NAD^+ during fermentation.

Time spent:0 sec
Subject:
Biochemistry

QID:401211
System:
Metabolic Reactions

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